

HORIZONTALISM, STRUCTURALISM, LIQUIDITY PREFERENCE AND THE PRINCIPLE OF INCREASING RISK

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ABSTRACT

It is shown that the horizontal post-Keynesian view of credit-money endogeneity is reasonable rather than extreme. One can argue in favour of a horizontal credit-money supply curve while still taking into account the non-accommodating behaviour of the central bank, financial innovations, portfolio adjustments, liquidity preference theory, and the principle of increasing risk. The differences between horizontalists and other post-Keynesian monetary theorists are matters of emphasis rather than substantial differences of opinion. The only real topic of contention is the ultimate determinant of the base rate of interest.

I INTRODUCTION

It is no exaggeration to say that the concept of an endogenous supply of credit-money has now been widely accepted among non-orthodox economists in general and post-Keynesian economists in particular. What money or credit endogeneity exactly means, however, has been a source of controversy. The consensus that seems to emerge from many of the surveys that have recently appeared on the subject is that reasonable people would necessarily agree that the credit-money supply curve is upward sloping; a long list of authors who have supported this view can usually be found in the footnotes of these review articles.¹ The opposite view is usually dispatched with one or two names, usually that of Basil Moore. The consensus seems to be that the few heterodox economists who argue in favour of a horizontal credit-money supply curve are holding overly radical views, which are unnecessarily extreme. In fact, it has

¹ Among these surveys, see Niggle (1991), Targetti (1992, p. 274), Dow (1994b), Hewitson (1995), Musella and Panico (1995). Davidson (1989, p. 490) refers to Moore's and Kaldor's viewpoint as an 'extreme form' of endogenous money—an opinion shared by Wray (1992, p. 297) who notes that 'the Horizontalist position is an extreme endogenous money approach', while Messori (1991, p. 148), an Italian circuitist, thinks that 'the choice of an infinitely elastic credit supply and of exogenous rates of interest leads from bad to worse'. A balanced overview is however presented in the survey of Cottrell (1994a), where the horizontal position is called 'the radical endogeneity theory'.

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been argued by Musella and Panico (1995) that those who cling on to this extreme view are repeating the errors of Keynesians of the past—illustrated by the Radcliffe report—giving to outsiders the erroneous impression that monetary matters are unimportant within post-Keynesian theory.

The purpose of the present paper is to show that this apparently consensual view is mistaken: there is also a relatively large group of economists who could be described as horizontalists;² horizontalism is a reasonable method to describe non-neoclassical monetary production economies; and horizontalism is fully compatible with an adequate representation of a generalized theory of liquidity preference or of the Kaleckian principle of increasing risk.³ The argument which I want to make is that neither of these two (relevant) concepts constitute appropriate foundations to claim that the LM curve and the supply curve of credit-money are generally upward-sloping.

It is obvious that the IS/LM model remains intact when endogenous money with an upward-sloping supply curve is introduced in the model. Whether the money supply is fully exogenous, or whether the money supply is endogenous but its curve of an upward-sloping shape makes no difference whatsoever. This sort of money endogeneity offers nothing new to differentiate neoclassical from post-Keynesian economics.⁴ In particular, the main result of the IS/LM model still holds: an increase in the level of output, following a shift of the investment schedule, *necessarily* leads to an increase in the rate of interest. This was precisely Keynes's main objection to Hicks: 'An increase in the inducement to invest *need* not raise the rate of interest' (Keynes, xiv, p. 80).

If the link between the interest rate and the level of output is not broken, some of the standard beliefs imbedded in the neoclassical paradigm are recovered: an increase in investment leads to an increase in the rate of interest, as the loanable funds theory would predict. We are back to the neoclassical

² Besides Moore, the following names are a sample: Arestis (1988), Arestis and Eichner (1988), Eichner (1986; 1987, ch. 12), Forman, Groves and Eichner (1985), Hicks (1982, ch. 19), Howells and Bain (1983), Kaldor (1982, 1983), Lavoie (1984, 1992, ch. 4), Le Bourva (1953, 1959, 1992), Rogers (1989, ch. 10), Smithin (1994a, 1994b), and several of the French circuit economists.

³ I took this view some ten years ago, as is acknowledged by Dow and Dow (1989, p. 163), who include an article of mine (Lavoie, 1985a) among the few studies which reconcile liquidity preference theory with the endogenous money approach. In other work (Lavoie, 1986, 1992), I have maintained that the concepts of liquidity preference, increasing risk and financial fragility need not be abandoned when taking a horizontalist stand. This has not deterred critics from claiming that I had discarded these concepts as unrealistic because they were providing support for a rising LM curve (Kregel, 1993, pp. 1315–6).

⁴ Hicks (1937, p. 157) himself made a case for the endogeneity of money within the framework of his newly-born IS/LM model, as the following quote clearly demonstrates: 'Only we can now generalise our LL curve a little. Instead of assuming, as before, that the supply of money is given, we can assume that there is a given monetary system—that up to a point, but only up to a point, monetary authorities will prefer to create new money rather than allow interest rates to rise. Such a generalised LL curve will then slope upwards only gradually—the elasticity of the curve depending on the elasticity of the monetary system (in the ordinary monetary sense)'. Within this context, it is no surprise that Kaldor (1937) would represent Pigou's model as one where the money supply increases when interest rates increase, and that a couple of years later Kaldor (1939) would graphically depict such a rising money supply curve.

world of scarcity, with crowding out effects and the like. For post-Keynesian monetary theory to be truly distinct from various neoclassical versions of monetary theory, both the loanable funds approach and the Wicksellian natural rate of interest must be discarded. While neither is sufficient, or probably necessary, this is *most simply* done by claiming that the central bank is in a position to set nominal rates—and even *real* rates of interest—by adjusting the former to the actual or expected inflation rate (Moore, 1988a, p. 265; Ciccone, 1990, p. 455; Pivetti, 1990, p. 460; Smithin, 1994a, p. 172).⁵ What the rate of interest exactly hinges upon is not really important provided it does not necessarily increase with the level (or the rate of growth) of economic activity.⁶ As Pasinetti (1974, p. 47) points out, what is required is that the rate of interest be ‘determined *exogenously* with respect to the income generation process. Whether, in particular, liquidity preference, or anything else determines it, is entirely immaterial’.⁷ It is simplest to proceed by assuming that the rate of interest is exogenous.⁸

By contrast, two main mechanisms have been offered by post-Keynesians to explain the upward-sloping money supply curve.⁹ The first mechanism is related to the degree of accommodation of the monetary authorities. Interest rates rise in expansion because the central bank refuses to accommodate the needs of the financial system. Interest rates may rise directly, following changes in the discount rate for instance; or interest rates rise because, following the lack of accommodation, the cost of funds for banks, on their liability side, is increased. The second mechanism can assume that the central bank accommodates, and is associated with the general notion of liquidity. The argument can be subdivided into three: interest rates rise because banks become illiquid; interest rates rise because their customers become less liquid; or interest rates rise because uncertainty rises in the view of the banks and the public. We shall analyze each of these mechanisms in turn.

But before we do so, two comments are in order. First, it will quickly become apparent to the reader that the paper is based on an implicit hypothesis, i.e., differences between various versions of post-Keynesian monetary analyses rest on shaded differences of emphasis, rather than on trenchant differences of substance. Perhaps the only real disagreement rests on my contention that the upward-sloping credit-money supply curve, just like the upward-sloping supply curve of output, must be forsaken if post-Keynesians are to free themselves from the stranglehold of orthodox neoclassical economics.

⁵For instance, Cottrell (1994b) argues that Moore’s horizontalist position is the most logical objection against the Fisher effect from a post-Keynesian standpoint.

⁶And indeed some post-Keynesian models with endogenously-determined interest rates offer such examples. In Palley (1991–92, p. 171), an increase in output induces a fall in interest rates. In Jarsulic (1989), there are periods where accelerating accumulation is accompanied by falling interest rates.

⁷In a meeting that took place on October 21st 1985, and during which I took notes, Kaldor expressed exactly this idea.

⁸It is also coherent with the behaviour of monetary institutions, as described by several practitioners (Goodhart, 1989, p. 581).

⁹It will quickly become apparent, as Dow (1994b, p. 17) says, that ‘Minsky (1975, 1982) has provided much of the inspiration for these developments’. See also Minsky (1986).

Secondly, there is some confusion in the literature as to whether one is dealing with money supply curves or with credit supply curves. Since post-Keynesians usually argue that causality runs from loans to money, it makes more sense to deal with credit supply curves.¹⁰ However, reflecting the ambiguities of past literature, the term credit-money will generally be used.

II LACK OF ACCOMMODATION

A set of exogenous base rates

While many post-Keynesians are tempted to accept the horizontal viewpoint for pedagogical reasons, they are often reluctant to do so because the horizontal money supply curve gives the impression that money only plays a passive role in the economy, or because such a position will leave them open to the obvious following criticism: if money is so fully endogenous, how can monetarism be a scourge?¹¹ Indeed, this is the most common complaint of the lack-of-accommodation critics of horizontalism.

For horizontalists the *base* interest rate is not a market phenomenon: it is a bureaucratically determined price, which may be more or less influenced by the political class and the financial lobby.¹² The non-elected bureaucrats of the central bank have discretionary control over this interest rate.¹³ While the central bank only indirectly influences the monetary aggregates, it can fix the base interest rate with absolute precision, within the boundaries of existing conventional financial wisdom. The base interest rate is an administered price, analogous to the prices set by the managers of oligopolies selling produced commodities.¹⁴ As a result the money supply curve, in symmetry to the supply-offer curve for the goods supplied by megacorporations, is best seen as a horizontal curve.

¹⁰ This point was brought to my attention by Peter Howells. Le Bourva, in a seminar in Rennes in the Spring of 1994, also pointed out that he wished he had used graphs in the credit/interest space (instead of money/interest space) in his 1959 article (translated in Le Bourva, 1992).

¹¹ This was a question which Harrington (1983) immediately addressed to Kaldor. A similar question was put to me by Gedeon (1985). My answer was that: '... Monetary authorities do have the opportunity and the ability to influence the real economy. In fact they have a lot of power over real output, as the recent past can testify' (1985a, p. 846); 'An endogenous money stock, with reverse causation, if it implies that money does not matter, does not imply that monetary policy is of no significance.... Monetary policy is a potent tool when drastic steps are taken. But it is also a far too dangerous one, as the recent past can testify...' (1991, p. 274).

¹² The base rate of interest is the rate under the control of the central bank. Depending on the institutions, this may be the discount rate, the overnight rate, the federal funds rate, the Treasury bill rate. The base rate may depend on the inflationary fears of the central bank, exchange rate considerations, and so on.

¹³ Since Keynes is usually considered to be the ultimate reference on those matters, one might as well quote him in his latest 1945 pronouncement on the issue: 'The monetary authorities can have any rate of interest they like.... They can make both the short and long-term [rate] whatever they like, or rather what they feel to be right....' (xxvii, pp. 390–1).

¹⁴ See Eichner (1987, p. 858). Moore (1991, p. 406) also uses the same image: 'So long as borrowers are within their credit allocation (overdraft limits), a horizontal market period supply function may simply be viewed as a graphic representation that the supply price is given in the market period, until sellers alter their administered price'.

This, of course, does not imply that megacorps never change their prices or are unable to do so. Similarly, to describe the money supply curve as a flat curve does not imply that central banks forever peg the base interest rate (Moore, 1988b, p. 384). It simply means that there are no natural forces in the economy, meaning no supply and demand market mechanism, that should compel interest rates to rise when economic activity rises. When the stock of money expands, rates do not have to rise, they may also decrease.¹⁵

Central banks can change the exogenous rate of interest whenever it pleases them to do so. Indeed, we may consider that each level of interest rate corresponds to a particular monetary regime or monetary policy. Graphically, each monetary regime may be represented by a different horizontal credit-money supply curve. This is clearly indicated by Kaldor, in a lucid footnote which has been omitted both by those who have attempted to delete him from the group of economists who support horizontalism,¹⁶ and by those who should have made good use of it. After presenting an algebraic representation of endogenous money, where the rate of interest is a given, Kaldor adds:

Diagrammatically, the difference in the presentation of the supply and demand for money, is that in the original version (with M exogenous) the supply of money is represented by a vertical line, in the new version by a horizontal line, or a set of horizontal lines, representing different stances of monetary policy (Kaldor, 1983, p. 22).¹⁷

A reaction function of the central bank

The notion among post-Keynesians that a lack of accommodation by the monetary authorities may jeopardize the full endogeneity of the money supply may be attributed to Chick (1977, ch. 5). She argued that central banks may act

¹⁵ An analogy can also be found with respect to pricing by firms. Lee (1994, p. 320) argues that factual evidence shows that when demand increases, prices or mark-ups need not rise: they may also fall or remain unchanged.

¹⁶ See Musella and Panico (1993) and Targetti (1992, ch. 11).

¹⁷ Emphasis added. What Kaldor says in the above quote is quite compatible with what he later wrote in a 1985 letter to Targetti, when asked to clarify his position (Kaldor, 1982, p. 24) about the horizontal money supply at a fixed interest rate: 'I do not want to claim that monetary policy is ineffective or that it does not have an influence on the interest rate. The diagram ... only entails that for every rate of interest the money supply is infinitely elastic' (Targetti 1992, p. 276). Now Targetti discards this straightforward answer, commenting that 'this explanation strikes me as elliptical and not entirely convincing'. Similarly, Musella and Panico (1993, p. 56) claim that Kaldor's horizontal money supply curve is only 'a simplifying analytical device'. If Kaldor had meant to imply that endogenous money is best described by the upward-sloping money supply curve that he had himself proposed in 1939, he would surely have said so. Instead he reasserts that 'the money supply is infinitely elastic' and relies on 'a set of horizontal lines'. In answer to a question regarding the possible misleading aspect of horizontalism, Thirwall turns out to be of an opinion similar to mine. According to Thirwall (1994, p. 77), Kaldor argues that 'at whatever the rate of interest the monetary authorities want to maintain, the supply of money is elastic with respect to demand, so it's completely horizontal at whatever the rate of interest that is set. The schedule moves up and down at the price at which the monetary authorities want to maintain the rate of interest. In that sense, he's a horizontalist'.

passively or may pursue dynamic operations. Since then, several authors, e.g. Rousseas (1989, p. 478), have argued that because an accommodating monetary policy is unlikely, a flat credit-money supply curve may only be considered as a simplifying pedagogical device. In a more sophisticated framework, the money supply curve should be drawn as an upward-sloping curve. Besides Kaldor's (1939, p. 14), the first such *graphical* representation is Rousseas's (1986, p. 94), where the rising money supply curve originates from a gradually shifting *vertical* money supply curve.

Rather than making use of a set of vertical supply curves, horizontalists would capture this upward-sloping curve by positing the existence of a set of *horizontal* supply curves. In the specific case where the central bank is leaning against the wind, there will be a set of temporally-ordered horizontal lines which, *when money demand is taken into consideration*, will constitute a positively-sloped money supply curve. *The upward-sloping curve is thus a special case, based on a particular feedback rule, of the more general horizontal depiction.* The simplest representation of this feedback rule is the one offered by Musella and Panico (1993, p. 58), where the rate of interest i set by the central bank depends on the differential between a demand-determined stock of money M and a targeted money supply M^* . In such a model, a shift in the demand for money will generate an increase in the interest rate set by the central bank, i.e., accommodation is only partial.¹⁸

$$i = i_0 + \eta(M - M^*)$$

More complex representations of the lean-against-the-wind reaction function can be constructed. It can be assumed, as does Palley (1994), that commercial banks stand ready to lend as much as desired at a given rate of interest—the lending rate of interest—set by a mark-up over the base rate.¹⁹ The base rate itself could depend on a mechanism similar to the one shown in the above equation. Whatever the exact algebraic representation, the following considerations arise.

While at every moment of time the supply of credit-money is fully horizontal, shifts in the demand curve for loans induce the creation of a set of temporally-ordered horizontal supply curves of credit-money, because of the non-accommodating behaviour of the central bank. This set of horizontal supply curves yields a dynamic upward-sloping supply curve of loans, because any increase in loan demand generates an increase in the base rate, and hence for a given markup, an increase in the lending rate. It should be noted that the slope of the dynamic

¹⁸ Such a reaction function includes the case of the so-called frown costs, when commercial banks are forced to borrow reserves. Rising frown costs, and its consequent rising credit-money supply curve, are thus just another aspect of the non-accommodating behaviour of the central bank. To a given monetary stand should correspond given frown costs.

¹⁹ In contrast to what Hewitson (1995, p. 292) and others claim, it should be noted that Moore (1988a) does not assume a constant mark-up since in his view federal funds rate (the base rate) varies continuously while lending rates are administered prices and vary discontinuously. The mark-up in his view thus continuously changes. It follows that empirical data showing the variability or the flexibility of the mark-up over base rates only disproves Rousseas's (1986) version of interest determination. It has no bearing on Moore's theory or the validity of horizontalism.

supply curve of credit-money depends on the slope and the intercept of the demand curve for credit-money.²⁰ Thus supply and demand curves are interdependent, a fact that was pinpointed before by Moore (1988a, p. 265): 'The long-run money supply function [the dynamic supply curve] is in general strictly undefined, since it is not independent of credit and money demand forces'.

It should also be clear that the upward slope of the dynamic supply curve rests on the non-accommodating behaviour of the central bank. Had the central bank pegged the federal funds rate, the dynamic supply curve of credit-money would be perfectly flat. If the central bank had decided to decrease the federal funds rate, the dynamic credit-money supply curve would be downward-sloping. Indeed, throughout the 1980s such an instance of a downward-sloping supply curve occurred, when the Fed continuously reduced interest rates while output and the money supply were expanding.

There is no serious disagreement on these issues any more. On the one hand Moore himself has acknowledged the existence of this long-term upward-sloping supply curve, when central banks are non-accommodating.²¹ On the other hand, authors more inclined to believe in the generality of the non-accommodating case have conceded the existence of a set of temporally-ordered horizontal lines.²²

Structural endogeneity

Now there are more complex variations on this non-accommodating theme, variations which try to explain what happens when demand for base money does not entirely create its own supply. This is where the so-called *structural view* of endogenous money comes into play (Pollin 1991). This view may be linked to the earlier Keynesian arguments based on the variability of the velocity of money. Various authors (Tobin, Rousseas, Minsky, Kaldor) asserted that a lack of accommodation by the monetary authorities would be countered by changes in the composition of portfolios and by financial innovations, in order to save on reserve requirements or cash requirements.²³ This also implies rising money

²⁰ This feature is not clearly illustrated by Palley (1994, p. 75).

²¹ 'If a central bank policy goal were to target the rate of growth of some particular monetary aggregate, interest rates would rise whenever income and the money stock increased. One could then imagine some longer-term "money supply function" upward-sloping with respect to interest rates... Even with such a policy regime in force the short-run money supply function would always be horizontal' (Moore, 1988a, p. 265).

²² 'Suppose the Fed is targeting the federal funds rate and has a "leaning against the wind" feedback rule so that, when bank lending increases, it raises the funds rate. In this case, at any moment the static loan supply schedule is horizontal, but the dynamic loan supply schedule is positively sloped' (Palley, 1991, p. 400); 'So the money supply has stopped being horizontal, or, if you like, has gone up to a new horizontal line... Now Vicky Chick has expressed that, I think, as follows: Basil Moore's lines are a series of horizontal lines behind an upward-sloping supply curve for money' (Reynolds, 1994, p. 135).

²³ This, as is recalled by Musella and Panico (1993), was already Kaldor's position in his 1939 paper. These authors thus opposed the Quantity theory of money on the basis of money endogeneity, not so much by invoking reverse causality as was to be done later, but rather by relying on a variable velocity of money. As I pointed out earlier, 'provided it is recognized that money does not determine income, post Keynesians can feel comfortable with either stable or unstable velocity' (Lavoie, 1985a, p. 845).

multipliers, or rather, falling *credit divisors*.²⁴ Under the structural view, the efficient restriction of monetary aggregates is a difficult task, but it will eventually hinder the normal behaviour of banking and financial institutions, create some inconvenience to them or their customers, and as a consequence lending rates will rise, even when there is no change in the base rate.

Why mark-ups on the base rate eventually rise has been explained in particular by Seccareccia (1988, pp. 56–8). He argues that as banks extend more credit in a non-accommodating situation, they are forced to lure deposits with lower reserve requirements. These term deposits are less liquid and therefore less attractive in the eyes of the public. Higher interest rates on term deposits or improved facilities for these deposits might thus be offered to depositors, to induce them to modify the structure of their deposit balances. The liabilities of the banks become more heavily weighted towards term deposits which carry higher rates of interest than demand deposits. The rising cost of funds on the liability side leads to increases in the lending rate of interest.²⁵ According to Seccareccia, but also to Pollin (1991, p. 375), it is the changing structure of the liability side of the balance-sheet of banks—rather than that of the asset side—which is responsible for the upward-sloping supply curve of credit. What must be noted is that the composition of the stock of money changes as a result of the non-accommodating behaviour of the central bank; it is not an endogenous feature of the system.

Structuralism is perfectly compatible with horizontalism, as long as one does not postulate that central banks always lean against the wind.²⁶ Nothing in the structuralist position contradicts horizontalism. In particular, it was never denied by horizontalists that higher interest rates would generate attempts to economize on cash or demand deposits.²⁷ These portfolio adjustments, however important they may be, rank second to the expansion of balance sheets required by economic expansion—the income effects highlighted by horizontalists.

²⁴ I have shown this in greater detail in Lavoie (1992, pp. 203–215).

²⁵ Here we have a full-cost theory of lending rates, where the level of the prime lending rate is an administered price based on the bank's average cost of funds.

²⁶ Black and Dowd (1994) give an instance of structuralism where higher aggregate demand yields lower rates of interest. Their LM curve is downward-sloping. A similar result is presented by Harcourt (1980, p. 157), when discussing the findings of the Radcliffe Commission and the work of Gurley and Shaw. Similarly Targetti (1992, pp. 273–281) gives a graphical depiction of structural accommodation, where anything can happen to interest rates when output increases.

²⁷ In contrast to what Palley sometimes suggests, no horizontalist would dispute that, whenever the pegged base rate changes, 'such changes will set up incentives for banks to reconfigure their balance sheet' (Palley, 1994, p. 79). In my own case, more than ten years ago, I wrote: 'If such [liquidity] pressures were exercised, the various economic units would use all sort of subterfuges to avoid the utilization of money (currency or checking accounts). There would be a move from banking activities toward non-banking financial activities, as explained by John Gurley and Edward Shaw: Banks would encourage the transformation of demand deposits for term deposits; large firms would start acting as banking institutions and credits between companies would be extended.... As a consequence the velocity of money as defined by the authorities would be on the rise' (Lavoie, 1984, p. 779). Elsewhere (*ibid*, p. 788), I stressed how changes in base interest rates induce changes in the composition of the liabilities of the banks. I also linked the so-called New View of Tobin and Teigen, which inspires most of Palley's analysis, to an unstable money multiplier and hence to a non-accommodating central bank (*ibid*, pp. 792–3).

III CHANGES IN OBJECTIVE LIQUIDITY RATIOS

Quantitative constraints and credit-worthy borrowers

Let us now abandon the determinants of the base interest rate, and look at the determinants of other rates, short or long, in particular the prime lending rate. The view being held by the proponents of the so-called liquidity preference theory is that even when base money or reserves are forthcoming without any restrictions at a constant base interest rate, it is very likely that increased activity will induce credit restrictions and rising lending interest rates, the latter being shown here on Figure 1. The following two quotes, by two leading exponents of this viewpoint, are quite representative of this opinion:

[An] increase in liquidity preference can nevertheless affect expenditure plans and outcomes, even when the monetary authorities stand ready to provide banks with all their reserve needs at a particular interest rate (Dow and Dow, 1989, p. 151).

The money supply curve is upward-sloping relative to interest rates in the absence of innovations or norms of behavior, even though there is no strict quantity constraint on bank lending.... It is the 'liquidity preference' of the banks which causes the short term interest to rise (Wray, 1990, pp. 166–7).

With these two quotes we find the two major ingredients of the so-called liquidity preference view: the rising credit-money supply curve and the non-accommodating behaviour of commercial banks (in contrast to that of the central bank). The quantitative restrictions which commercial banks impose on their customers is thought by some authors to be a subject of dissension among post-Keynesians. The reader is sometimes given the impression that those who support horizontalism are unaware of those quantitative constraints.²⁸ Some

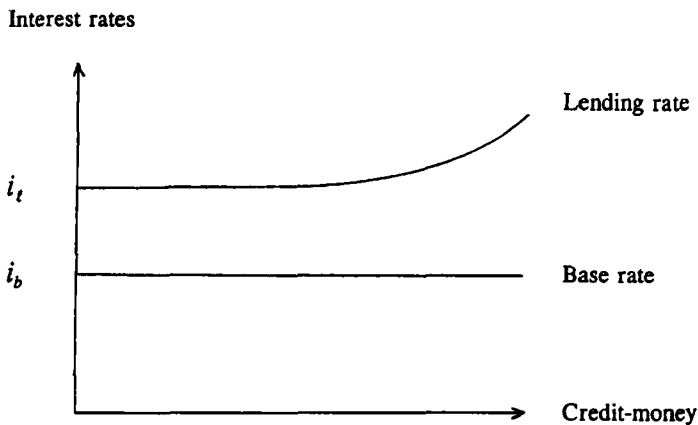


Figure 1. Rising lending rates despite a pegged base rate

²⁸ Despite his friendly review of horizontalism, Cottrell (1994a, p. 599) writes: 'On the Kaldor–Moore view ... the banks are mere ciphers in this process, passively accommodating whatever demands they happen to experience'.

post-Keynesians object to Moore's *non-discretionary* lending behaviour of the banks, arguing that banks often choose not to lend.²⁹ The claim, quite legitimate, that banks have some restrictions on their lending does not call into question the validity of horizontalism. The price of loans—the interest rate—is an inadequate exclusion mechanism. Because of uncertainty and the complexities of decision making, bank lending depends on whether or not the potential borrower fulfils various norms and customs. 'The amount of advances the banks can make is limited by the demand from good borrowers' (Robinson, 1952, p. 29).³⁰ The norms establishing what a good borrower is, change with the circumstances. Indeed all the main advocates of horizontalism have underlined the presence of these norms.

At any one time the volume of bank lending or its rate of expansion is limited only by the availability of credit-worthy borrowers. When trade prospects are good or when the money value of borrowers' assets (collateral) rises as a result of a rise of prices, the demand for bank credit rises but by the same token the credit-worthiness of potential borrowers also improves, so that the demand for and the supply of credit move simultaneously in the same direction (Kaldor, 1981, p. 15).

Commercial bank loan officers must ensure that loan requests meet the bank's income and asset collateral requirements. They must in general satisfy themselves as to the credit-worthiness of the project and the character of the borrower. It is precisely for these reasons that banks develop client relationships with their borrowers (Moore, 1988a, p. 24).³¹

On this point, there is no difference between the proponents of horizontalism and the other advocates of money supply endogeneity.³² One may thus conclude

²⁹ See Gedeon (1985), Messori (1991), Musella and Panico (1993), Dow (1994a). It is true that Moore (1988a, ch. 2–3) sometimes pushes the non-discretionary argument a bit too far, as is shown by some of the above authors. But this is strictly a matter of emphasis. One may wish to highlight either the availability of overdraft facilities or the existence of a fringe of unsatisfied borrowers. Proper emphasis may depend on the stage of the business cycle.

³⁰ This is repeated later: '[A] fall in activity sharp enough to cause losses deprives the banks of credit-worthy borrowers and brings a contraction in their position' (Robinson, 1970, p. 510).

³¹ The issue has recently been further clarified by Moore (1994, p. 123). Having reiterated that banks grant 'credit limits according to their estimates of borrowers' credit worthiness, and then are price setters and quantity takers in their retail loan and deposit markets', Moore warns that 'this is not to deny that many small borrowers are effectively credit-constrained. New businesses and poor households in particular do not possess the income, assets, and credit records criteria that banks require in order to make profitable and financially sound loans (the banks' three C's: credit, collateral and character)'.

³² This can be seen by comparing the quote of the previous footnote with the the following quote: 'Up to some point, determined by conventions regarding acceptable balance sheets, banks will increase the money supply without raising interest rates on loans. This does not mean, of course, that banks will meet all loan demand—many potential borrowers will not obtain loans even if they are willing to pay the market interest rate, since banks ration credit based on rules of thumb, risks assessment, established relations with customers, and other conventions of behavior' (Wray, 1991, p. 15).

that all post-Keynesians are aware of the constraints that could be imposed upon would-be borrowers. This is *not* a point at stake.³³

The liquidity ratios of borrowers and Kalecki's principle of increasing risk

The issue rather is to find the best way to represent the availability of credit-worthy borrowers. One graphical way, suggested by Hewitson (1995), is to assume that the interest rate on loans rises as the debt to equity ratio of the customer (the firm) rises. This is shown in Figure 2, and it corresponds to Kalecki's 1937 principle of increasing risk (or to Keynes's borrower's and lender's risks). The higher the debt to equity ratio, the more risky the borrower, and hence the less likely the borrower will remain credit-worthy. What we have here is a cross-section of firms at a moment in time. If there are two otherwise identical firms, A and B, with different leverage ratios, one is expected to be charged a higher rate of interest than the other. This is surely a graph we can all live with. It corresponds to the practices of rating companies, as is done by Standard and Poor's or Moody's for instance: more risky companies are able to borrow, but at higher interest rates due to their lower rating.

Those who argue that the credit-money supply curve is upward sloping because of the illiquidity of banks' customers need to take one *very crucial* step further however. As noted by Hewitson (1995, p. 296) in an important footnote to her graph, they need to make the *additional* assumption that the debt to equity ratio rises with increased borrowing. This is a reasonable assumption to make when dealing with a single firm taken in isolation. But is it a relevant assumption at the *macroeconomic* level? The curve in Figure 2 now needs to be interpreted as the illustration of a time-series, with the whole economy, on

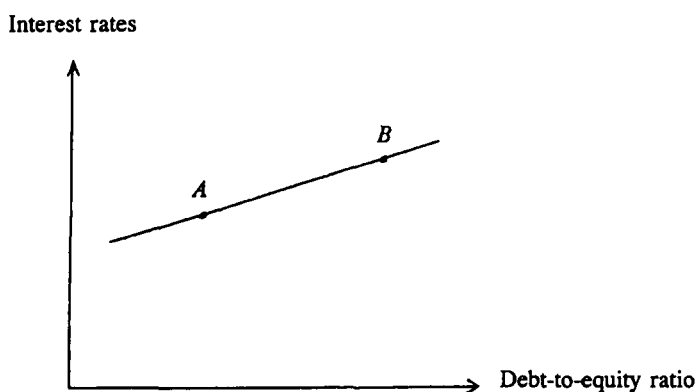


Figure 2. Liquidity: Cross-section or time-series?

³³In a reply to a comment by Gedeon (1985), I myself made the following clarification: 'What I said, or perhaps what I meant, was that the central bank was ready to provide all required credit at the going rate of interest to all financial entrepreneurs, while commercial banks were ready to do the same for non-financial entrepreneurs, under the proviso that the respectability of the borrowing party or of its project could be established' (Lavoie, 1985a, p. 844).

average, moving from A to B in historical time. This is a decisive assumption to make. Does the leverage ratio of industrial firms rise when economic activity increases? The belief that it does correspond to Minsky's financial fragility hypothesis. According to Minsky (1975, ch. 5,6), as the boom develops, the acceptable leverage level may rise and hence firms will take more risks and more leveraged positions. Minsky (1975, p. 112) illustrates this contention by drawing his famous graphical representation of investment decisions based on borrower's risk and lender's risk. As investment accelerates, firms would switch from conservative hedge finance to risky speculative finance.

Minsky's financial hypothesis may however be challenged from both the historical and the theoretical point of view. With respect to history, Isenberg (1988), for instance, has shown that the debt ratios of manufacturing firms in the 1920s diminished rather than increased during the boom preceding the Great Depression.³⁴ Factual evidence then need not be favourable to Minsky's thesis. A casual look at the trend in leverage ratios over the last two recessions does not seem to bring support to his thesis. Procyclical debt to equity ratios have *not* been a characteristic of *each* business cycle.³⁵

With respect to theory, the case of the representative firm developed by Minsky does not necessarily carry over to the macroeconomic setting. When investment increases, profits increase as well, unless other elements induce reduced profits, such as a higher rate of savings by households or a deficient trade balance. Assuming away these external factors, the realized leverage ratio is just as likely to fall, although entrepreneurs and their bankers are willing to increase the debt ratio. Individual efforts to increase leverage ratios may lead to lower aggregate debt to equity ratios, while any effort to reduce leverage ratios may lead to higher actual leverage ratios. This *paradox of debt* is occasionally recognized by Minsky, for instance when he says that 'the improvement of realized profits partially frustrates the planned debt-financing of investments of firms' (1975, p. 114).

There are of course strong similarities between Minsky's arguments and those that are often attributed to Kalecki's earlier writings to justify a positive relationship between the level of income (or the rate of investment) and the rate of interest. The paradox of debt, which was extensively analyzed by Steindl (1952, pp. 113–121), himself a Kaleckian, calls into question the extension to macroeconomics of Kalecki's 1937 version of his principle of risk.³⁶ What

³⁴ Her more recent findings on this issue are mixed (Isenberg, 1994).

³⁵ An exponent of structural endogeneity thus makes the following devastating remarks: 'Despite the distinctiveness of the neoclassical and Minskian approaches, there is an important common error in both arguments, rendering both inadequate for explaining the patterns since the mid-1960s. If corporations were motivated to increase debt financing either because of borrowing costs or as a result of a boom psychology, we would then also expect this increase in debt financing to be accompanied by an increase in corporate spending.... But in fact what has accompanied the rise in corporate debt financing since the mid-1960s has been a decline in investment growth' (Pollin, 1987, p. 151).

³⁶ This was pointed out informally in Lavoie (1983). I have more recently put forth a growth model operating on the principle of effective demand, where higher rates of accumulation may be accompanied by falling realized leverage ratios (Lavoie, 1995a). In some recent models of Minsky's economics (Keen, 1995), the realized rate of profit (and hence the leverage ratio) is still impervious to effective demand.

must be concluded on this issue is that neither Kalecki's principle of increasing risk nor Minsky's financial fragility hypothesis, whatever their virtues within their domain of validity, can sustain the hypothetic construction of an upward-sloping credit-money supply curve.³⁷

This is further reaffirmed when we look at the way that firms obtain new loans. Established firms usually have lines of credit which largely exceed their needs. Any decision on the part of the firm to draw on its credit line would be automatically made good, at the ruling prime rate of interest. Such loans are thus automatically created, without a discretionary decision on the part of the managers of the bank. The principle of increasing risk, as initially presented by Kalecki (1937), does not quite hold for the analysis at hand. Credit will not be granted to the individual firm at an increasing cost by the banking sector. A firm can draw as much as it wants on its agreed line of credit, at a fixed cost. The cost of borrowing is set in advance, according to the risk grade assigned to the firm, which may depend more on the absolute size of the firm's earning assets than on any measure of its debt to equity ratio.³⁸ This corresponds to Le Bourva's (1992, p. 448) analysis, when he says that banks fix 'a set of rates', each rate corresponding to the assessed risk class, and then lend 'however much borrowers ask for, *provided that they can offer satisfactory collaterals*'.³⁹

We close this section by coming back to the question of the best representation of these quantitative constraints. How are we to represent them, if the credit-money supply curve is best viewed as horizontal? The answer that is offered here is similar to the one given by Davidson (1972, p. 72). When discussing the determinants of the prices of existing capital goods, Davidson introduces a *shift variable* in the demand function. This shift variable, noted E, 'represents the number of entrepreneurial investors who can obtain finance for their demand for capital goods'. The rules of thumb used as financial norms by the bank managers are thus shift variables which have nothing to do with the slopes of the demand for and supply of credit (Lavoie, 1985a, p. 845; Arestis and Eichner, 1988, p. 1010). The relevant demand for credit should be its *solvent* demand.⁴⁰ We may, if we want, consider the demand for credit curve to represent the demand for credit arising from credit-worthy borrowers, i.e., those borrowers who fulfil the norms imposed by the banking system. More exacting norms will shift the solvent demand for credit curve to the left. We may also

³⁷ It should be noted that Kalecki's equation of profits is first taken into account by Minsky only in 1977! When Minsky (1982, pp. 231–257) deals with a macroeconomic relation to explain debt ratios, he does so within the neoclassical context of a loanable funds approach. See Lavoie (1995b).

³⁸ This is recognized by Dow (1994a, p. 17), when she indicates that risk assessment is in an inverse relation to the amount of relevant information, which is particularly lacking for small firms.

³⁹ Elsewhere, he writes that 'All other things being equal, the credit extended by the banker to one client is certainly not infinitely elastic; as Kalecki had already emphasized, the banker's growing risk limits the size of the loan which he or she would be willing to grant' (Le Bourva, 1992, p. 459).

⁴⁰ I have myself utilized the expression 'effective demand of entrepreneurs' (Lavoie, 1992, p. 177), and so has Ülgen (1995), but I believe the term 'solvent demand' is more appropriate and less ambiguous.

want to represent the limits on credit lines by a truncated supply of credit: beyond some point, the supply simply *vanishes*.

The liquidity ratios of banks

Leaving aside the liquidity of the banks' customers, let us now consider the balance sheet of the banks themselves to see if they could justify an upward-sloping credit-money supply curve. This is a position which can be found in Dow and Dow (1989) and which has been mainly presented by Wray:

Given a degree of liquidity preference, as banks balance sheets expand, perceived lender's and borrower's risk rises so that the banks raise retail loan rates of interest. In consequence the money supply curve is upward sloping with respect to the interest rate (Wray, 1992, p. 305).

Those who take such a position can easily rely on the writings of Keynes, for it is Keynes (xiv, p. 219) himself who sums up his liquidity preference theory by claiming that the rate of interest is determined by the interplay between the liquidity preference of the public and that of the banking system. He adds that the supply curve of money may be rising in the interest rate space as it 'depends on the terms on which the banks are prepared to become more or less liquid' (*ibid*, p. 221). Keynes believed that an increased demand for initial finance cannot be met without a rise in the rate of interest, unless the banks are ready to augment loans at the current rate of interest. According to Wray, when banks make loans they check their own level of liquidity, that is the ratio of loans to equity and the ratio of loans to safe assets such as government bonds and reserves. Wray and others believe that these ratios rise during an economic expansion, and hence, 'even where additional loans can be made which are no more risky than loans which have been made previously, banks may require higher interest rates to compensate for riskier leverage ratios' (Wray, 1990, p. 179). Thus even if borrowers' risk stays constant, lenders' risk will eventually rise.

Let us suppose, for the sake of discussion, that banks carefully keep track of their liquidity ratios, however they are measured.⁴¹ As was done with borrowers in the previous section, one has to distinguish what is true for a single bank from what is true for the banks taken as a whole, i.e., a valid concept in a microeconomic situation may not be relevant in a macroeconomic setting. If a single bank is pursuing aggressive lending policies, as it could in the present environment of generalized liability management, its credit-worthiness may become questionable and other financial institutions may refuse to lend to it. The main objective limit to the expansion of a bank is the rate of expansion of the other banks and the norms of prudent and acceptable behaviour self

⁴¹ Note however that commercial banks operating in pure overdraft economies own no liquid assets. They simply borrow the funds that they require. How is bank liquidity to be measured in those financial systems where liability management is a structural feature? This may explain why capital adequacy requirements (equity ratios) are now being put in place by most central banks.

imposed by the banking industry (Wray, 1990, p. 182). This is just another application of Kalecki's principle of increasing risk. And indeed, Dow and Earl (1982, p. 140) provide a graphical representation of that principle applied to banks, based on Minsky's investment graph. In their graph, a shift of credit demand generates greater lenders' risk and higher debt to bank equity ratios.⁴²

This graph, however, just like Minsky's graph, is built on a *ceteris paribus* assumption, whereas the supply of funds for a single bank should depend on what the other banks are doing. It may turn out that all the other banks are also generously creating credit, thus creating an additional supply of funds for any given bank, and allowing all economic units to make more profits. If banks have a higher rate of return on their loans than on their safe assets, the value of their equity could grow at a fast pace, so that the loans to equity ratio need not rise during the expansion. But let us suppose, again for the sake of argument, that loans to bank equity ratios rise in the expansion.⁴³ Does it entail higher rates of interest on loans?

It has been argued by many authors that when banks 'move forward in step', they can safely create all the credit money that is required for production to increase (Keynes, v, p. 23). There is no theoretical limit to the amount of credit money which, overall, the banking system can create to satisfy the requirements of increased activity, provided the central bank helps to accommodate. If *all* banks, in contrast to one bank, pursue aggressive lending policies, the behaviour of none of them will appear questionable (Le Bourva, 1992, pp. 459–461).⁴⁴ The liquidity of a bank in modern financial systems depends 'upon the extent of the funds which it thinks itself (and is thought by those who trade with it) to be able to borrow' (Hicks, 1974, p. 54). The norms to follow are the average leverage and liquidity ratios of all the other banks of the same size. If all banks are expanding in step, all banks will experience similar transformations of their balance-sheet. Indeed, this is why norms change for structural reasons. If it happens that the liquidity ratios of all banks fall during expansion, banks will not necessarily attach more risks to these lower ratios.⁴⁵ The objective fragility of the banking system may be greater but, as will be pointed out in the next section, its subjective assessment by bankers might not change.

⁴²Incidentally, in that graph, Dow and Earl turn out to be horizontalists, by Dow's own definition, since the lending rate is exogenous and constant whatever the risks involved. This is noted by Wulwick (1984, p. 345).

⁴³Wray (1995, p. 276) claims to provide such a proof. He does so essentially by assuming that during an expansion highly-leveraged banks grow faster than low-leveraged ones. But this contradicts his belief, shared by all, 'that no bank can afford to grow at a rate greatly in excess of the growth rates of other banks' (*ibid.*, p. 277). Using a temporal argument which is reminiscent of Asimakopulos's sequential logic, Wray also points out that at the very moment a new loan is granted, the loan to equity ratio must rise, since equity is given at that moment of time. This cannot be disputed. But as Asimakopulos (1991, 113) points out, 'banks may not have any particular reason to want their initial liquidity position restored before they extend new loans. They grow and prosper by becoming less liquid, by increasing their loans, as long as interest is always earned on these loans'.

⁴⁴In Le Bourva (1993, p. 218), the validity of Kalecki's principle of increasing risk in a microeconomic setting is also contrasted to its lack of relevance within a macroeconomic context.

⁴⁵In the end, this crucial point is acknowledged by Wray (1995, p. 279).

Some authors, for instance Dow and Dow (1989, p. 164) and Jarsulic (1989, p. 38), have linked the upward-sloping credit-money supply curve to the empirical evidence gathered on the importance of the *degree of liquidity pressure* (Arestis and Driver, 1988; Forman *et al.*, 1985; Eichner, 1986). This variable is broadly defined as the ratio of total loans to total deposits, as can be found in the balance sheet of banks. What has been shown is that increases in the degree of liquidity pressure, compared to its trend value, induce higher interest rates and a slowdown in lending. For Dow and Dow, this shows that the balance sheet position of banks, i.e., their liquidity, is influential in determining the interest rate at which they will lend and their willingness to do so. If one assumes further that increases in economic activity always lead to more illiquid balance sheets, then the upward-sloping credit-money supply curve would be demonstrated on those grounds.

But as Dow and Dow (1989, p. 164) recognize, this is not Eichner's interpretation of his findings. According to Eichner (1986, p. 106), fluctuations of the degree of liquidity pressure, compared to its ever changing trend value, are due to the non-accommodating behaviour of the central bank. When the central bank does not accommodate, commercial banks are forced to sell their Treasury bills to the public; the deposits of the public will shrink as a result of the efforts of banks to reduce their secondary reserves to acquire primary ones. Banks may also be forced to borrow funds from other financial institutions, thus diminishing the proportion of deposits among their liabilities. In both cases, the ratio of loans to deposits will rise. The increases in the degree of liquidity pressure and in interest rates are caused by a third factor: the non-accommodating behaviour of the central bank. We are back to the non-accommodating rising money supply curve that was previously identified as a special case. It cannot be claimed that those who support horizontalism are unaware of the issues related to the liquidity ratios of banks. Eichner, who attached so much empirical importance to the degree of liquidity pressure, remained a staunch exponent of horizontalism (Eichner, 1987, p. 859).

IV A GENERALIZED VIEW OF LIQUIDITY PREFERENCE

Liquidity preference and animal spirits

It has been claimed by proponents of liquidity preference theory that there is no incompatibility between such a theory and the endogenous money approach (Wray, 1990, p. 170; Dow and Dow, 1989, p. 163). Dow (1994b, p. 6) adds that this compatibility between liquidity preference and money endogeneity holds as long as credit-money is seen to be constrained to some degree. This would seem to include horizontalism, having been shown in the sub-section above headed *Quantitative constraints and credit-worthy borrowers* that the main advocates of horizontalism recognize that banks set quantity constraints. But Dow excludes horizontalism by adding that compatibility requires the money supply curve to have 'upward slope with respect to the interest rate'. My claim is that

there is no incompatibility between the principle of liquidity preference and a horizontalist approach to endogenous credit-money.⁴⁶

In Section III above, a rather narrow view of liquidity has been taken. Liquidity was defined as some ratio, some indicator of a firm's ability to fulfil its obligations. This sort of liquidity is mainly associated with the objective notion of risk. Risk assessment and credit-worthiness, while relying on objective liquidity ratios, are also a largely subjective matter. It is to these more subjective aspects that we now turn.⁴⁷ Post-Keynesians have associated the concept of liquidity preference with an environment of fundamental uncertainty. Within that environment, economic agents do not know probability distributions, they are aware that past sequences of events may not reoccur, they hope to learn more in the near future, and they may therefore delay decisions. This sort of liquidity preference is inversely related to the degree of confidence, to the reliability of beliefs, to Keynes's weight of arguments and animal spirits (Runde, 1994). A generalized theory of liquidity preference is thus intertwined with a generalized theory of rational behaviour in an environment dominated by uncertainty. Reise (1992, pp. 294–5) for instance links Keynes's animal spirits to the liquidity preference of commercial bankers. I have myself underlined the importance of this interpretation of liquidity preference from the very beginning, in the case of banks, linking it to rational behaviour when animal spirits falter, and so has Moore.⁴⁸

When bankers begin losing some of their high 'animal spirits', though they are aware of the fact that their new behavior will harm the economy, they prefer to restrain their creation of credit-money. They know that those banks that are the least affected by the recession are those banks that show the most moderation. For this reason, it is quite possible for the banking system to start reducing its credit lines just when firms need extended loans (Lavoie, 1984, p. 791).

An improvement in long-run investor expectations will lead banks to extend credit to those borrowers whose projects now appear more attractive.... The elasticity of the supply of credit to demand may change over the business cycle if the willingness and the capacity of financial institutions to supply credit changes.... The price and availability of credit play a critical role in the business cycle.... In contractions, the real cost of credit may rise just when the demand for liquidity is at its most urgent ... (Moore, 1988a, pp. 145–6).

⁴⁶ Dow (1994b, pp. 6–7) implicitly admits it when she writes: 'Suppose Keynes had chosen instead to assume a given interest rate rather than a given money supply. Would this have implied an unlimited willingness on the part of the banks and the monetary authorities to meet increased demand for liquidity? Clearly... Keynes would not have taken this view'. It is therefore possible to describe the credit-money curve as horizontal and take the view that banks may be more or less receptive to credit demand.

⁴⁷ Hewitson (1994) makes this distinction between objective factors, related to Kalecki's principle of increasing risk, and subjective factors, related to liquidity preference *per se*.

⁴⁸ These statements fit the distinction made by Wray (1991, p. 15; 1992, p. 303) between a demand for credit-money and a demand for liquidity: banks will increase credit-money when they share the optimism of entrepreneurs, but they will decline to do so when this increase in credit-money arises from rising liquidity preference or faltering animal spirits.

It is in this sense that liquidity preference and the theory of effective demand are the two sides of the same coin (Kregel, 1984/5, p. 142). Most proponents of an endogenous approach to money would agree that increases in economic activity usually accompany conditions which are conducive to reduced liquidity preference. The risk of default, whatever the objective conditions, is perceived to be lower.⁴⁹ On these grounds Wray grants that interest rates may not rise in an expansion, despite the rising leverage ratios of banks or of their borrowers. But Wray insists that this statement is conditional on a *fall* in liquidity preference. Since everyone agrees that such a fall is likely in an expansion, besides the semantics there is not much left to disagree about.

The upswing is thus characterized by a rising demand for credit to finance new projects, a demand which is accommodated with enthusiasm by the banks and by wealth-holders, domestic and foreign. This process reflects, and is facilitated by, the fall in liquidity preference which results from increasing confidence in high returns on non-money assets (Dow and Dow, 1989, p. 159).

Given a state of liquidity preference, expansion of balance sheets, leveraging of capital and reserves ... can occur only at rising interest rates. However if liquidity preference falls, balance sheets can expand without pressure on interest rates. This is why Keynes said that expansion *might, but need not* raise interest rates. It depends on liquidity preference (Wray, 1995, p. 278).

The 'low animal spirits' of horizontalists are thus equivalent to the 'high liquidity preference' of other endogenous money theorists. Both concepts relate to issues of confidence and expectations. The focus of those using the notion of animal spirits is usually on the expansion of output, sales and credit aggregates, while that of those favouring liquidity preference is on interest rates and asset prices. Emphasis is different, but the intent is analogous.⁵⁰ Since matters of confidence may affect the behaviour of all agents, whether they be banks, firms or consumers, an appropriately generalized theory of liquidity preference will take all these actors into consideration (Le Héron, 1986; Dow and Dow, 1989). Within that framework, a world of ultimate liquidity preference is a world where firms would refuse to produce for fear of indebtedness, where banks would refuse to lend for fear of loan defaults, and where consumers would refuse to spend for fear of unemployment (Lavoie, 1983, p. 323).

⁴⁹ Indeed, empirical evidence shows that risk premia are lower not higher in expansions (Juttner 1994, p. 503; cf. Robinson 1956, p. 230). This is consistent with the statement, made in the sub-section above headed *The liquidity ratios of borrowers and Kalecki's principle of increasing risk* that borrowers are encouraged to increase their leverage.

⁵⁰ Kaldor, who is sometimes accused of having unnecessarily discarded liquidity preference, having called it a 'red herring' (Kaldor, 1982, p. 26), has also put forth views which are like the generalized liquidity preference approach favoured here. In the passage already quoted, he says that 'when trade prospects are good ... the demand for bank credit rises but by the same token the creditworthiness of potential borrowers also improves' (1981, p. 15). Similarly I have written that 'in periods of recession or of high uncertainty, fewer borrowers or/and fewer projects are considered worthwhile' (Lavoie, 1985a, p. 844).

Liquidity preference and interest rate differentials

In less radical terms, agents might simply be looking for safe assets rather than unsafe ones, or for clearly defined short-term assets rather than long-term assets the return of which is uncertain. Briefly put, a generalised liquidity preference theory tells us to what extent the various agents in the economy are ready to become illiquid and to abandon liquid assets, as can be found in Keynes's *Treatise on Money*. We are far beyond the standard interpretation of liquidity preference—more akin to Keynes's *General Theory*, with its precautionary and speculative balances—which is associated with a simple choice between bonds and cash, or with the demand for money function. This neoclassical standard interpretation is not what post-Keynesian authors have in mind, as they have pointed out on a number of occasions.

Again, the question arises about how best to represent liquidity preference. Various answers may be offered, depending on the aspects that need to be emphasized. Several authors have suggested, following Hicks (1974, p. 45), that broadly speaking liquidity preference is measured by the spread between short and long interest rates.

... Liquidity preference determines not the long rate of interest but the spread between the short and the long rates... (Wells, 1983, p. 533).

Liquidity preference, then, in practice determines the difference between the interest rate on liquid deposits and on less liquid substitutes. The monetary authorities set the rate at the short-term end of the spectrum; liquidity preference (along with other considerations) determines the mark-up to long term.... An interest rate set by the monetary authorities is consistent with varying rates on bank loans, depending on the state of liquidity preference (Dow and Dow, 1989, pp. 148, 158).

The suggestion offered by Dow and Dow corresponds precisely to what has been maintained here all along.⁵¹ The central bank determines the base rate, and all other rates are adjusted to that rate, through liquidity preference or other considerations. Liquidity preference does not determine *the* rate of interest,⁵² as we were led to believe by the narrow interpretation of liquidity preference offered by the standard IS/LM model. Rather liquidity preference determines the *differential* between the base rate and all the other rates. Looking more

⁵¹ Indeed, in a letter sent to Rousseas in April 1985, following the reading of a paper of his which objected to both Moore's horizontalism and Kregel's (1984–85) liquidity preference theory, I commented that 'I agree by and large with [a view which holds that] for a given stance from the central bank, several interest rates and loan levels are possible, depending on the liquidity preference of banks'.

⁵² This is what I meant when I wrote, thus producing some apparent irritation among fellow post-Keynesians (Wray, 1990, p. 156; Dow, 1994b, p. 20), that 'Keynes' liquidity preference theory is well known. Post-Keynesians do not abide by it anymore, however' (Lavoie, 1985b, p. 76). I believe it is also this restricted interpretation of liquidity preference that Kaldor had in mind when he disparaged it. Indeed, in the meeting of 21st October 1985 already alluded to, Kaldor indicated that the lending rate is influenced by the bankers's liquidity preference, which the central bank tries to modify.

specifically at the liquidity preference of banks, it could be argued that it is measured by the differential between the base rate and the prime lending rate (Lavoie, 1992, p. 195).

With this sort of interpretation, there is no need for proponents of horizontalism to disregard the role played by liquidity preference. In particular, there is no need to argue that the long rate of interest is only determined by expected future short rates (Moore, 1988a, ch. 11; 1989, 487). Mark-ups may vary quite a bit for reasons related to liquidity preference, and expectations about future rates may only be part of the 'other considerations' that help to determine rate differentials, but this need not jeopardize the horizontalist view. That there is no serious incompatibility between the diverse views being held should be obvious from the way prime lending rates are said to be set: both sides deny that they result from a simple and rigid mark-up rule; both believe that they result from a discretionary decision of bank managers. What is the basis of this decision is a matter of emphasis.⁵³

Liquidity preference and the base rate

A possible source of tension between horizontalists and other post-Keynesians is the determination of the base rate itself, and possibly its practical relevance for the economy. It could be argued that liquidity preference determines the base rate, where liquidity preference is defined as all the factors which govern the monetary policy of monetary authorities. It could then be contended that *market opinion*, or the financial lobby, determines the liquidity preference of the central bank, and hence the conventional rate that the central bank wishes to establish.⁵⁴ This said, it remains that in my view and that of Moore (1991, p. 412), 'central banks must be held accountable for the ruling level of short-term interest rates, no matter how much they disavow'.

But even if this is taken for granted, what remains at stake is whether the short-term base interest rate set by the monetary authorities will play any role in the economy. Is the market continuously able to short-circuit the monetary policy of the central bank? On the one hand, those who have emphasized the importance of liquidity preference may be tempted to answer yes. *The* rate of interest relevant to the economy, then, would not be the base rate but rather the long-term rate, which could durably pursue a trend of its own. This would be a revised instance of the liquidity trap, where the central bank

⁵³ For Wray (1992, p. 305), the mark-up would mainly depend on the perception of lenders's risk, while for Moore (1988b, p. 382), it would mainly depend on the costs of intermediation and a target rate of return on equity. If risks are perceived to be larger, it should presumably be reflected in a higher target rate of return gross of default losses, and hence in a larger mark-up.

⁵⁴ For instance, Pollin (1991) shows that market-determined long-term rates of interest Granger-cause federal funds rate which presumably are under the ultimate responsibility of the monetary authorities. According to Rogers (1989, p. 232), the rate of interest 'is set by convention—the balance of power between the rentiers and the capitalists as mediated by the monetary authorities'. A similar position is that the rate of interest 'is the result of a convention ... a consensus between those who can manipulate and those who can benefit from interest rates' (Lavoie, 1985b, p. 78).

would manage to get the base rate down, but without any impact on other interest rates.⁵⁵

Horizontalists on the other hand, according to which the relevant rate is the base rate, would rather answer negatively: the market cannot continuously short-circuit monetary policy. If monetary authorities are sufficiently insistent and consistent, a shift in interest rate differentials can only be temporary. Summoning Keynes's authority, it could be said that the convention established in the prevailing spread 'will not be always unduly resistant to a modest measure of persistence and consistency of purpose by the monetary authority' (vii, p. 204). In the case where long term rates would be high relative to short rates, financial operators would come to realize that substantial profits can be made by borrowing short and lending long. Eventually, unless 'the authorities' nerves are shaken by the ferocious growls with which the bears have been deafening them all this time', the convention enshrined in the base rate should prevail, and the liquidity premium should be back to its normal level (Robinson, 1952, p. 30).

Ultimately, what might be at stake here is a theory relating rates of profit to rates of interest in the long period. For those who believe in the strong influence of market opinion, the realized or expected rate of profit governs the rate of interest over the long haul. Now the logic of the whole scheme emerges. When the economy is accelerating, growth rates are higher, and so become realized or expected profit rates (along the lines of the Cambridge equation). The equalization of asset rates of return induces higher long-term interest rates, and the effect is eventually propagated to other market rates.⁵⁶ All this is compatible with a rising credit-money supply curve, where the latter is reinterpreted in growth terms. In contrast, for those who believe in the ability of the central bank to eventually impose market rates of its choosing, the rate of interest determines the normal rate of profit, just as Sraffa (1960, p. 33) expressed it in his famous sentence, and just as Keynes would have it in chapter 17 of the *General Theory*.⁵⁷ It is precisely in this chapter that Keynes discards the Wicksellian natural rate of interest, thus making the (real) interest rate free to be set permanently at whatever level the monetary authorities see it fit to be.⁵⁸

⁵⁵ This seems consistent with the following: 'One problem with Keynes's explanation of the liquidity trap is that it obviously cannot apply to short-term rates.... Although never explicitly stated, his liquidity preference theory must refer only to long-term interest rates' (Moore, 1988a, p. 247).

⁵⁶ 'Thus in the long run... the rate of interest must be governed by the rate of profit or the state of long-term expectations. In the absence of uncertainty, the interest rate would be pushed up to the point where it just equalled the rate of profit.... In the long run, the interest rate is determined by the state of long-term expectations—that is, by expectations of profit on real assets' (Wray, 1991, pp. 16, 18). Similarly, according to Nell (1988), the causality runs from the first to the last variable in the following way: rate of accumulation, realized rate of profit, normal rate of profit, normal long-term rate of interest.

⁵⁷ See Lavoie (1992, p. 361), Moore (1988, pp. 245–6), Panico (1985), Parguez (1975, p. 208) and Pivetti (1990). The adjustment process follows the full-cost principle—lower interest costs reduce target rates of return, and hence the prices of produced goods. It does not rely on changes in the cost of production based upon the marginalist principle of diminishing returns, as in Keynes (1936). What will happen to the realized rate of profit is another matter.

⁵⁸ As Smithin (1994b, pp. 156–7) puts it: 'In the absence of a natural rate of interest, it can also be argued that central bank control over short real rates will ultimately influence the entire structure of interest rates in the economy, including long rates.... Eventually, the real economy must adjust to the policy-determined interest rate, rather than vice-versa. This is therefore the precise opposite of the natural rate doctrine'

CONCLUSION

We may now recapitulate the main findings of the paper. I have tried to argue that credit-money supply should be viewed as a horizontal curve in the money and interest space, and hence that the LM curve, if any validity can be attached to it, should also be viewed as a horizontal curve. I have contended that this horizontalist position should not be considered to be extreme. Rather it should be seen as a consistent and logical position.

In a world where the Wicksellian natural rate of interest does not exist, the monetary authorities are free to fix the short rates, even real rates, at the level of their choice within wide limits. The decision to accommodate or to refuse to accommodate increases in credit-money demand should thus be viewed as a bureaucratic or political decision. Higher interest rates accompanying expanded activity are not a natural endogenous phenomenon. The upward-sloping credit-money supply curve should be seen as a special case, based on a non-accommodating feedback rule, of the more general horizontal depiction.

Besides non-accommodation by the central bank, non-accommodation by the commercial banks or the public has been the reason given to justify the upward-sloping credit-money supply curve. It has been shown that advocates of horizontalism have acknowledged that commercial banks constrain the creation of credit-money, by relying on criteria defining credit-worthiness, and that these criteria evolve with changing circumstances.

Three components of the liquidity preference view have been examined, based on the liquidity of the borrower, the liquidity of the banks and the general liquidity sentiment of all agents. It has been concluded, on the basis of empirical and theoretical considerations at the macroeconomic level, that these constructions could not carry the weight of justifying the contention that rising interest rates are *necessarily* associated with increased activity.

The main result of the present study is that a horizontalist position may be upheld, while simultaneously taking into account the principle of increasing risk and a generalized liquidity preference approach, involving fundamental uncertainty and animal spirits. The extent of the validity of those two principles must however be correctly ascertained. In particular, the liquidity preference of banks and of the public determines neither the base rate nor the bond rate; rather it determines the spread between the lending rate and the base rate, or between the bond rate and the base rate, the latter being set by the central bank.

One may conclude that if post-Keynesian authors have diverse opinions as to the shape of the credit-money supply curve or as to the determinants of the structure of interest rates, these disagreements are mainly the result of differences in *emphasis*. The ultimate determinant of the *base* interest rate may remain the only serious source of contention; but even on this issue, the gap between the various views may turn out to be rather narrow.

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